

PAPER 1 / QUESTION 55

8822-6501 / original question 25

- 55.** Water is to be pumped through a vertical height difference of 12.5 m. The pump is driven by a wind turbine that has an efficiency of 50 % and an area swept by the blades of 100 m^2 . The average wind speed is 5.0 m s^{-1} and the air density is 1.2 kg m^{-3} .

What is the maximum mass of water that can be pumped every second?

- A. 3 kg
- B. 30 kg
- C. 60 kg
- D. 120 kg